

The E0 pair channel in ${}^3\text{He}(n, \cdot){}^4\text{He}$

Broken into two parts — *which* excited state we form, and *how* it decays — to find a second, γ -dark source of e^+e^- pairs

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction; physics, numbers and text reviewed by D. Neff.

Part 0 — Summary

The problem splits cleanly into two parts, linked by the spin-parity J^π of the intermediate ${}^4\text{He}^*$ state:

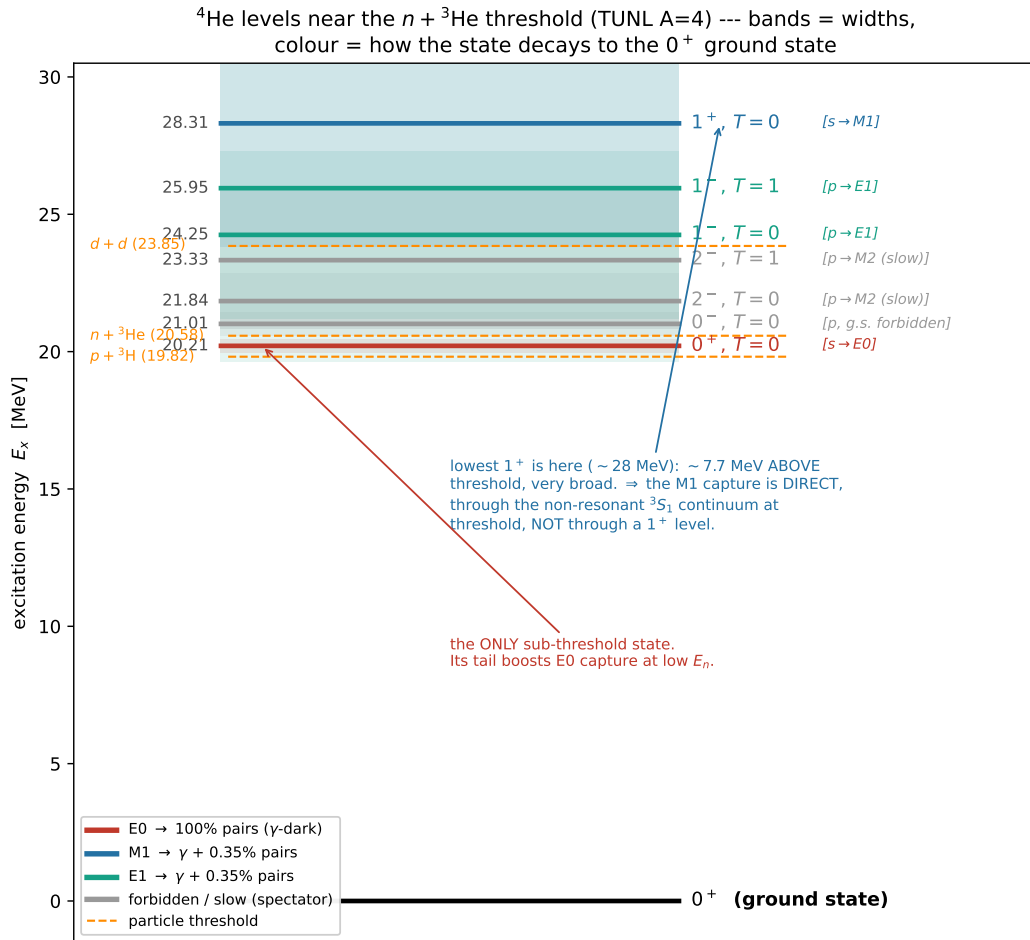
- **Part 1 — Formation:** which ${}^4\text{He}^*$ state does the captured neutron make, vs energy? Set by angular momentum (s-wave $\rightarrow 0^+, 1^+$; p-wave $\rightarrow 0^-, 2^-, 1^-$). **This sets the energy *shape*.**
- **Part 2 — Decay:** given that state, how does it reach the 0^+ ground state and does it make a pair? Set by selection rules. **This sets the *height* (pair fraction).**

Key results: (i) Only *three* states make ground-state pairs: 0^+ (E0, **100 % pairs**, γ -dark), 1^+ (M1) and 1^- (E1, both 0.35 % pairs). 0^- and 2^- are spectators. (ii) A $0^+ \rightarrow 0^+$ transition can go *only* by E0 — not M1 (the $0 \rightarrow 0$ rule forbids both the photon and M1). The M1 is a *different* capture channel. (iii) **There is no low-lying 1^+ level** — the lowest is ~ 28 MeV, so the M1 capture is *direct* (non-resonant) through the continuum; the 0^+ is the only state with a (sub-threshold) resonance helping it. (iv) The ${}^4\text{He}$ levels are so broad ($\Gamma \gtrsim |E_r|$) that capture is **mostly direct/continuum**, not sharp resonant capture — except for a real, broad sub-threshold boost of the E0 at low E_n ($\sim 10\times$). (v) The E0 and M1/E1 pairs sit at *opposite* ends of the energy range (E0 low, M1/E1 MeV); the absolute E0 size is the one unknown, $f = \sigma_{\text{E0}}/\sigma_{\text{M1}} \sim 10^{-3}\text{--}10^{-2}$.

Part 1 — Formation: which ${}^4\text{He}^*$ state do we make?

A slow neutron and ${}^3\text{He}$ fuse into ${}^4\text{He}^*$ at ≈ 20.6 MeV. Which excited state forms depends on energy; *whether it makes a pair* is decided later (Part 2).

The states that exist — and the 1^+ surprise



The ^4He levels near the $n + ^3\text{He}$ threshold (20.58 MeV), from the TUNL $A = 4$ evaluation [1]. Two things to read off:

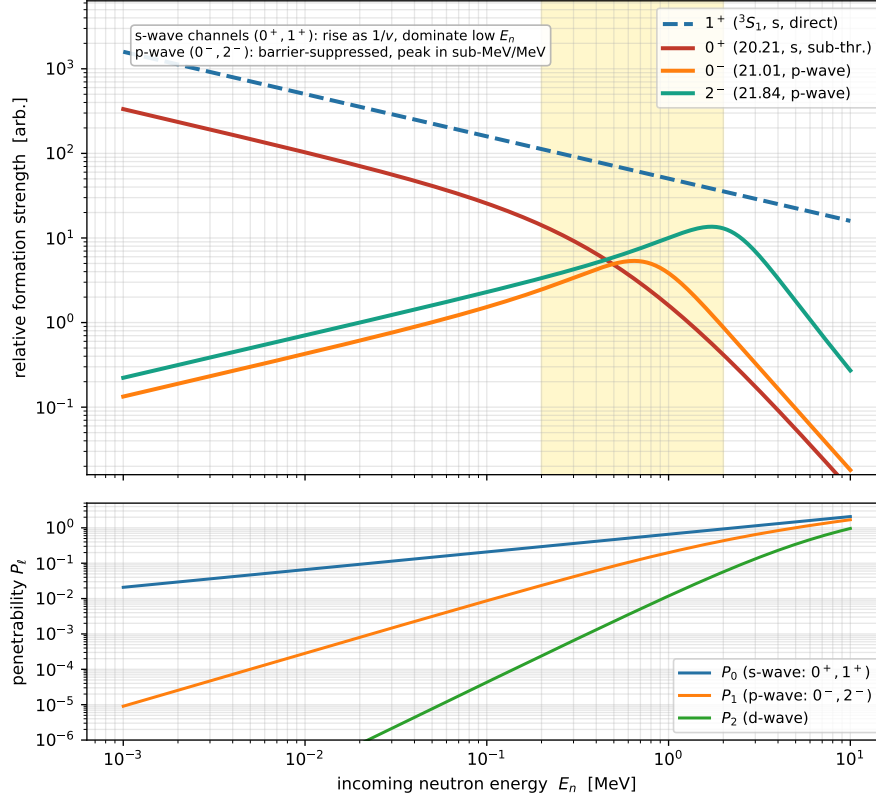
- The levels are **broad and overlapping** (widths 0.5–13 MeV) — ^4He has no sharp levels above threshold.
- **There is no low-lying 1^+ state.** The lowest 1^+ is at ~28.3 MeV — about 7.7 MeV *above* threshold and very broad. (This is the answer to “where is the 1^+ ?” — it simply isn’t down here.)

Which state, and when: angular momentum

Parity fixes the entrance orbital angular momentum, $\pi = (-1)^\ell$:

reached by	$\rightarrow$ states	energy behaviour
s-wave ($\ell = 0$)	$0^+, 1^+$	no barrier — rise as $1/v$, dominate at low E_n
p-wave ($\ell = 1$)	$0^-, 2^-, 1^-$	barrier-suppressed — switch on at sub-MeV/MeV

Which ${}^4\text{He}^*$ doorway do we form, vs neutron energy?
(single-level Breit-Wigner $\times$ penetrability; TUNL $A=4$, $a = 4$ fm. 1^+ is direct/non-resonant.)



Both 0^+ and 1^+ (s-wave) dominate at low E_n ; 0^- and 2^- (p-wave) peak at $E_n \approx 0.6$ and ≈ 1.7 MeV — *not* forbidden at low energy, just barrier-suppressed. **As the neutron energy drops, we form relatively more 0^+ and 1^+ .**

Continuum (direct) vs resonance

Is this capture *resonant* (through a sharp level) or *direct* (into the continuum, the photon emitted during scattering)? A level acts as a sharp resonance only if its width is smaller than its distance from the capture window, $\Gamma < |E_r|$. For ${}^4\text{He}$ that is never the case (Fig. 1): every level has $\Gamma/|E_r| \approx 1.2$ – 1.9 , so **the capture is mostly direct/continuum capture**, modulated by broad structures, not sharp resonant capture. Two consequences matter for us:

- **1^+ (M1): pure direct.** Its nearest level is 7.7 MeV away and 9 MeV wide, so $g(E)$ is flat ($g_{\text{thr}}/g_{1\text{ MeV}} \approx 0.8$): the famous $54\mu\text{b}$ M1 capture is *direct continuum capture*, with no resonant enhancement. (This is why ${}^3\text{He}(n, \gamma)$ follows a clean $1/v$ with no thermal resonance.)
- **0^+ (E0): direct + a sub-threshold boost.** The 20.21 MeV 0^+ sits 0.37 MeV *below* threshold; its tail enhances the E0 capture by $\sim 10\times$ at threshold relative to 1 MeV — a real, but *broad*, resonant boost concentrated at the lowest E_n .
- $0^-, 2^-, 1^-$ (p-wave) peak in the sub-MeV/MeV region and only matter there.

Part 2 — Decay: what does each state do?

Whatever state forms, ${}^4\text{He}^*$ almost always just **breaks back up** ($p + {}^3\text{H}$, the (n,p)t reaction, or $n + {}^3\text{He}$) and makes no pair. The rare (10^{-8} – 10^{-4}) **electromagnetic drop to the 0^+ ground state** is the only pair source, and *what kind* of drop is fixed by the state's J^π . Taking each state on its own:

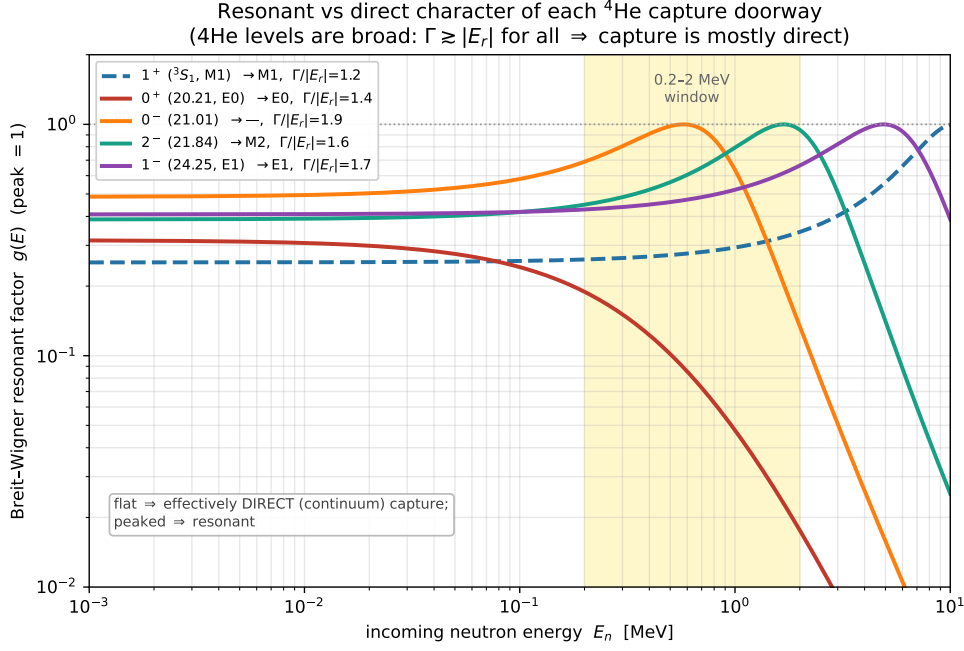


Figure 1: How “resonant” each doorway is across the capture window: the Breit–Wigner factor $g(E)$ (peak = 1). **Flat** $\Rightarrow$ **effectively direct** (continuum) capture; **peaked** $\Rightarrow$ **resonant**.

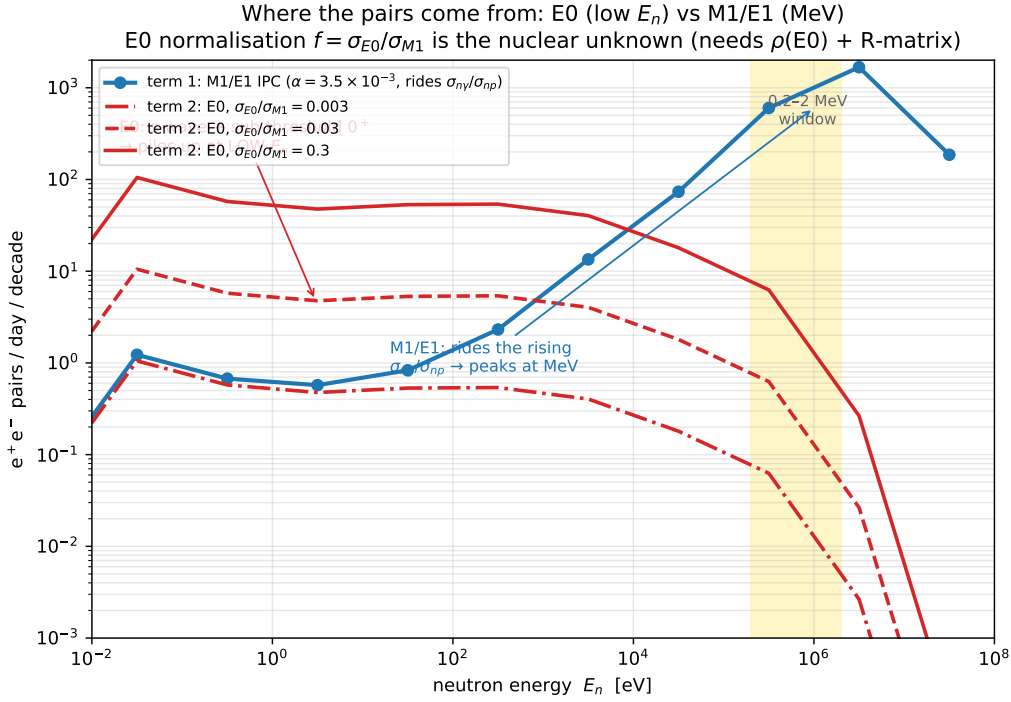
State	$\rightarrow 0^+$ g.s.	pairs?	role for our signal
0^+ (20.21)	E0 (γ forbidden)	100 %	E0 pair source — low E_n
1^+ (3S_1)	M1	0.35 %	M1 pair source — the ordinary $54 \mu\text{b}$
1^- (24–26)	E1	0.35 %	E1 pair source — rises toward MeV
0^- (21.01)	forbidden	—	spectator (breaks up)
2^- (21.84)	M2 (too slow)	~ 0	spectator

Row by row:

- $0^+ \rightarrow 0^+$: **E0**. A photon is forbidden ($0 \rightarrow 0$), so the *only* outlet is an e^+e^- pair — **100 % of these transitions are pairs**. This is the γ -dark channel, invisible to ENDF/Geant4 (and absent from our generator).
- $1^+ \rightarrow 0^+$: **M1**. Emits a 20.6 MeV γ ; $\sim 0.35\%$ of those convert to a pair (the internal-pair coefficient $\alpha_{\text{IPC}} \approx 3.5 \times 10^{-3}$ at this energy, anchored to the *measured* ^{12}C 15.1 MeV M1 value $(3.3 \pm 0.5) \times 10^{-3}$ [6]; BrIcc’s tables stop at 6 MeV [5]).
- $1^- \rightarrow 0^+$: **E1**. Same as M1 (a γ with a 0.35 % pair tail), but p-wave, so it grows toward MeV — this is what makes the radiative cross section rise (the giant dipole turning on).
- $0^- \rightarrow 0^+$: **nothing**. $0 \rightarrow 0$ forbids every photon multipole, and a parity flip forbids E0 too — so the 0^- has no electromagnetic route to the ground state and simply decays back to particles. A pure spectator (even though it peaks near 0.6 MeV).
- $2^- \rightarrow 0^+$: **M2**, far too slow at 20 MeV; the state breaks up first. Spectator.

Putting the two parts together

Part 1 sets the shape, Part 2 sets the height, and the total pair yield is the sum over the three source states, $N_{\text{pairs}}(E_n) = \sum_{J^\pi} [\text{formation}(J^\pi, E_n)] \times [\text{pair fraction}(J^\pi)]$. The two pair families end up at opposite ends of the energy range:



- **M1/E1 (blue):** 0.35 % pair fraction, but rides the rising radiative cross section $\Rightarrow$ **peaks at MeV**. This is the piece we can already compute (≈ 2200 pairs/day, 0.1–10 MeV).
- **E0 (red):** 100 % pair fraction *and* the low-energy s-wave + sub-threshold boost $\Rightarrow$ **piles up at thermal / sub-keV**.

So the sub-keV region we had written off (for M1 it is dead) might be exactly where the E0 pairs live — a genuinely different place to look.

The one unknown: the height of the E0 curve

Everything above is settled except *how strong* the E0 channel is, i.e. $f = \sigma_{E0}/\sigma_{M1}$ — physically, how often the 0^+ doorway makes its E0 transition to the ground state rather than breaking up. A literature dig found: the ${}^4\text{He } 0_2^+$ E0 strength *is* measured (the MAMI monopole form factor [2], the “ α -particle monopole puzzle” [3]) and is a real, collective transition — but that is the *decay-side* matrix element; turning it into a capture rate needs reaction theory. The ATOMKI ${}^4\text{He}$ pair work [4] models its pairs as E1/M1 capture, not a clean E0 line, and no calculation of ${}^3\text{He}(n, e^+e^-){}^4\text{He}$ exists. A transparent order-of-magnitude argument (statistical 1S_0 weight $\frac{1}{4}$ as a floor; E0 having no real-photon mode as a ceiling) brackets

$$\boxed{f = \sigma_{E0}/\sigma_{M1} \sim 10^{-3}\text{--}10^{-2}} \quad \Rightarrow \quad \frac{\text{E0 pairs}}{\text{M1 pairs}} = \frac{f}{\alpha_{\text{IPC}}} \sim 0.3\text{--}3$$

at low E_n — E0 plausibly *same-order-or-larger* than the M1 pairs in the sub-keV. The clean way to replace the bracket with a number is a ${}^4\text{He}$ multi-level R-matrix (Hale’s evaluated system) for the 1S_0 capture, normalised by the measured (e, e') strength. *Next step, worth doing together.*

What this sets up

With the shapes (Part 1) and pair fractions (Part 2) in hand, the remaining step is to fold in the absolute normalisations and the detector acceptance to get the final metric — X17 / IPC pairs per pulse (per day) vs neutron energy — now *including* the E0 channel. That is the next note.

Reproducing the figures

Scripts (in `scripts/`): levels — `make_he4_levels_fig.py`; doorways — `calc_doorway_formation.py`; direct-vs-resonant — `calc_direct_vs_resonant.py`; pair yield — `make_e0_pair_yield_fig.py`. Companion markdown (in `docs/e0-branch/`): the spine is `formation_and_decay.md`; see also `capture_breakdown.md`, `doorway_states_note.md`, `ipc_estimation_method.md`, and `ipc_roadmap.md`.

References

- [1] D. R. Tilley, H. R. Weller, G. M. Hale, “Energy levels of $A = 4$ ”, Nucl. Phys. A **541** (1992) 1 (TUNL); G. M. Hale’s ^4He R-matrix.
- [2] S. Kegel *et al.*, “Measurement of the α -Particle Monopole Transition Form Factor”, Phys. Rev. Lett. **130** (2023) 152502 (MAMI A1).
- [3] “Derivation of transition density from the observed $^4\text{He}(e, e')^4\text{He}(0_2^+)$ form factor ...”, arXiv:2306.07268; arXiv:2406.00653.
- [4] A. J. Krasznahorkay *et al.*, “New anomaly observed in ^4He ...”, Phys. Rev. C **104** (2021) 044003; arXiv:2104.10075.
- [5] T. Kibédi *et al.*, BrIcc, Nucl. Instrum. Meth. A **589** (2008) 202; <https://github.com/IAEA-NSDDNetwork/BrIcc> (pair coefficients tabulated only to 6 MeV).
- [6] Measured ^{12}C 15.1 MeV M1 IPC, $\alpha_\pi = (3.3 \pm 0.5) \times 10^{-3}$ (PEPSI; arXiv:nucl-ex/9704007).
- [7] M. Schlüter, G. Soff, W. Greiner, Z. Phys. A **286** (1978) 149; At. Data Nucl. Data Tables **24** (1979) 509 (high-energy IPC).
- [8] R. Wervelman *et al.*, $^3\text{He}(n, \gamma)^4\text{He}$ M1/E1 content, Nucl. Phys. A **526** (1991) 265.